BA820 – Project M4

Cover Page

- **Project Title:** Beyond the Timeline: Mapping the Evolution of Casting Norms and Romantic Age Gaps in Cinema
- **Section and Team Number:** B1 Team 10
- **Student Name:** Bhavya Bavishi

1. Refined Problem Statement & Focus (~0.5 page)

For M4, I keep the same substantive question but refine it into a robustness-focused individual analysis: Are the M3 findings about headline packaging and star-proxy patterns robust to reasonable specification choices? More specifically, I test whether the relationship (or lack of relationship) between within-film inequality packaging and star-proxy concentration remains stable under: different star-proxy definitions, temporal splits (early vs. late films), parity threshold changes (3 vs. 5 years), and bootstrap uncertainty assessment.

This is an upgrade and not a fresh direction. In M3, I was part of the interpretation that the use of headline couples (the biggest-gap couples in multi-couple films) is a practical method of investigating how inequality is bundled within films. In M4, I explicitly test the robustness and defendability of the star-proxy interpretation, rather than it being an artifact of a single proxy definition or a single sample split. This milestone contribution is thus methodological: I shift to a descriptive integrated result to a robustness-based confirmation of the packaging + star-proxy narrative, as I maintain the analysis in connection to the overall theme of the project, which is romantic age-gap inequality in cinema.

The M4, in contrast to M3, which rested on the cross-level integration between regimes, packaging, and the ecosystem of directors, isolates the director-ecosystem mechanism, and the internal structural aspects are tested. It becomes important not to show how level-level linkage can be established, but to assess the stability of ecosystem segmentation by directors under a variety of alternative specifications, such as having multiple random seeds, hierarchical clustering cross-checks, and time. This is a conceptual development of breadth of integration (M3) to depth of structural validation (M4).

2. EDA & Preprocessing: Updates (~0.5 page)

My M4 notebook is built **from scratch,** so it does not depend on the previous runtime state. To ensure consistency with the team's M3 workflow, I recreate the key derived tables used in the integrated analysis: 1. **Couple-level enriched table** and 2. **Multi-couple movie-level table with packaging metrics**

I recreate the M3-style **era labels** by clustering on standardized release_year and age_gap (KMeans with k = 3) so the enriched couple table remains structurally compatible with prior milestones, even though the main M4 focus is on the packaging and star-proxy layer. Next, I reconstruct the **movie-level packaging table** for films with at least two couples and compute the same interpretable packaging features used in M3: top_gap_dominance (largest gap − second largest gap), second_gap, median_gap, count_large, headline_share (largest gap / total gap in the movie). I then cluster these movies using KMeans (k = 3) into the same narrative-style packaging groups used in M3: **Headline-dominant, Recurrence-leaning, Low-gap / typical.** This reconstruction is important because all M4 robustness tests are performed on a clean, reproducible movie-level table built in the notebook itself, not imported from prior runs.

3. Analysis & Experiments (~2 pages)

This will be a milestone to continue my M3 contribution to the study to check the strength of the headline packaging + star-proxy relationship. The analysis is deliberate, and the techniques employed to analyze are course-related (clustering, rank correlation, nonparametric comparison, bootstrap uncertainty, and sensitivity checks) instead of using irrelevant models.

3.1 Rebuilding a Clean Headline-Couple Table

Determining the definition of the headline couple objectively was a major move in M3. In both movies, the couple whose age gap is the highest is the headline couple. In case there are several couples with the highest number of ties, I solve ties deterministically through a star proxy (maximum number of times a cipher is repeated). This is to avoid the null hypothesis (which was tested and disproved earlier) that couple number is a measure of narrative significance. These proxy variants are the main part of my M4 robustness design. In M2/M3, I made one single recurrence-based proxy; in M4,I am testing the fact that the conclusions are dependent on that specification.

3.2 Narrative of Baseline Reproduction of M3 Packaging.

I will later replicate a baseline summary by movie packaging cluster with the M4 movie-level analysis table (multi-couple films with headline-couple properties). Before carrying out robustness checks, I repeat the baseline summary by movie packaging cluster with the M4 movie-level analysis table. The following summary substantiates the same overall M3 structure:

- It is more dominant in terms of median dominance and median headline share in headline-dominated films.
- The largest cluster and are more distributed are recurrence-leaning films.
- Low-gap / typical films represent a smaller single group with less overall gap strength.

This measure is necessary since M4 does not intend to substitute M3; it seeks to determine whether the interpretation of M3 may be altered when assumptions change, taking place around the star proxy and thresholds.

3.3 Proxy Sensitivity: Stability in Correlations between Star-Proxies and Star-Proxies.

The initial robustness test is the investigation of whether the correlation between packaging concentration and star- proxy measures is a sensitive factor to the definition of proxies. I calculate the Spearman rank correlation of each of the three proxy variants with topgapdominance and headline_share. In all three proxy definitions, the correlations are extremely small and statistically unimportant (approach zero). This implies that a single arbitrary definition of the proxy does not lead to the M3 observation of low star- proxy correlation with packaging concentration. Rather, it seems an equilibrium outcome. It is a significant M4 result: it confirms the meaning that headline packaging and star recurrence are not tightly correlated in this data, at least in terms of the recurrence-based proxies that can be used in this case.

3.4 Cluster Comparison Test: Headline-Dominant and Other Films.

Secondly, I check the hypothesis that there is a systematic difference between the star-proxies distribution of headline-dominant and all other films. A Mann-Whitney U test (nonparametric) is used by me since:

- The groups are imbalanced,
- The distributional assumptions are not well stated,
- And the medians are more understandable than the means of counts of recurrence.

I do this comparison individually on all three star-proxy variations. The sample sizes suffice to make tests, but the headline-dominant films and other films have very similar medians, and the tests do not have strong evidence of separation. This finding once again favors the same direction as the correlation analysis: the packaging difference (headline-dominant and other) is evident structurally, but not well explained in this formulation by the recurrence-based star proxy.

3.5 Dynamic Strength: Early vs Late Films.

The possible weakness of pooled analysis is that the relationships can vary across periods. To monitor this, I divided the movie-level table into: early films and late films with median(release year) in the M4 movie-level sample (split year = 2005). This provides an almost equal split. Spearman correlations of packaging concentration and variants of star-proxies are recalculated within each period. The indicators and values do vary a little over time (as one would expect with smaller subsamples), but no stable and robust correlation is found in early or late movies. This indicates that the insignificant association between packaging concentration and the recurrence-based star proxy is not necessarily a weakness of the pooled-sample type; it is rather low even following a simple temporal separation.

3.6 Sensitivity or 3 vs 5 Years Parity Threshold

In M3, a strict threshold was used to determine near-parity (<= 3 years). In M4, my team-standard threshold is the same: I test a more liberal alternative (<= 5 years) to test the sensitivity. I calculate the number of headline couples that pass near parity under each of the two thresholds within the three movie packaging groups. The anticipated change is quite mechanical (the percentages of the near-parity levels improve after the 5-year mark), yet the overall cluster pattern does not change:

- Even headline-dominating films have very low headline parity,
- recurrence-leaning movies are more paritatory than headline-oriented movies,
- and the more inclusive story behind M3 is retained.

This implies that the understanding of the packaging is not sensitive to a small threshold change.

3.7 Bootstrap Uncertainty of Median Star-Proxy Difference

The sample size used is more than 100 and is sufficient to make conclusions about the entire population under study (Ryan 2005, p. 37). Lastly, I include a bootstrap uncertainty analysis of

the median difference between star-proxies values between:

- Headline-dominant films
- All other films

I bootstrap the median difference and provide a 95 percent interval of each Proxy variance. These intervals have zero and are reasonably broad in comparison with the estimated median difference (which is close to zero). This can be applied to interpretation since it does not over-read the null-like results. Instead of invalidating the effect categorically with the phrase no effect, the bootstrap framing justifies a more cautious statement:

- The differences in medians are slight, which have been observed.
- uncertainty is substantial,
- And the evidence is not that a strong or constant star-proxy separation by package cluster.

4. Findings & Interpretations (~1 page)

The sophisticated analysis reinforces the finding that director casting behavior sexual activity is an organized behavior as opposed to haphazard differences in motion pictures. Directors continuously cluster into separate ecosystems that are defined by age-gap tendencies and collaboration concentration. The observation that these clusters are very stable during repeated model runs indicates a meaningful structural differentiation of the segmentation and not algorithmic noise.

Among the major findings is the fact that age-gap conduct and the collaboration framework seem to be closely connected. Directors with a higher proportion of age gap in their movies also have a more focused collaboration structure, whereas those with more moderate age gaps have a larger actor base. This trend means that the age differences in love relations are not just solitary artistic decisions but can be supported by repetitive working relations and internalized production standards. In terms of industry, this implies that the results of representation can be caused by the existence of stable casting ecosystems, not by the decisions about the project that are made on an individual basis. Time is troped with temporal segmentation. Although the structure of the ecosystem is consistent throughout the career stages, a significant proportion of directors change cluster memberships between the early and late periods. This means that casting ecosystems are not fixed and are structured. This seems to be the behavior of some directors, as their casting priorities seem to be changing with time, possibly as a response to shifting audience expectations, industry demands, or personal artistic growth. Notably, movement is partially related to smaller segment-level samples, which implies that not every reclassification depicts a substantive behavior change.

Taken together, these results indicate that the issue of age-gap inequality in cinema has its roots in repetitive systems of collaboration, which are resistant to the modelling of variation and can, however, evolve. In the case of the studios, creative executives, and diversity programs, this means that the issue of representation might have to be addressed at the network level, rather than by a few movies.

Appendix

Shared GitHub Repository (Required)

- https://github.com/patelparisha/ba820-b1-team10
- Colab Link - https://github.com/patelparisha/ba820-b1-team10/blob/Milestones/Bhavya%20Bhavshi/M4_BhavyaBavishi.ipynb
- My files exist in the Bhavya Bavishi folder in the Milestones branch of the git repository.

Supplemental Material (Highly Recommended)

Figure 1. Silhouette Scores Across Cluster Specifications (k = 2–6)

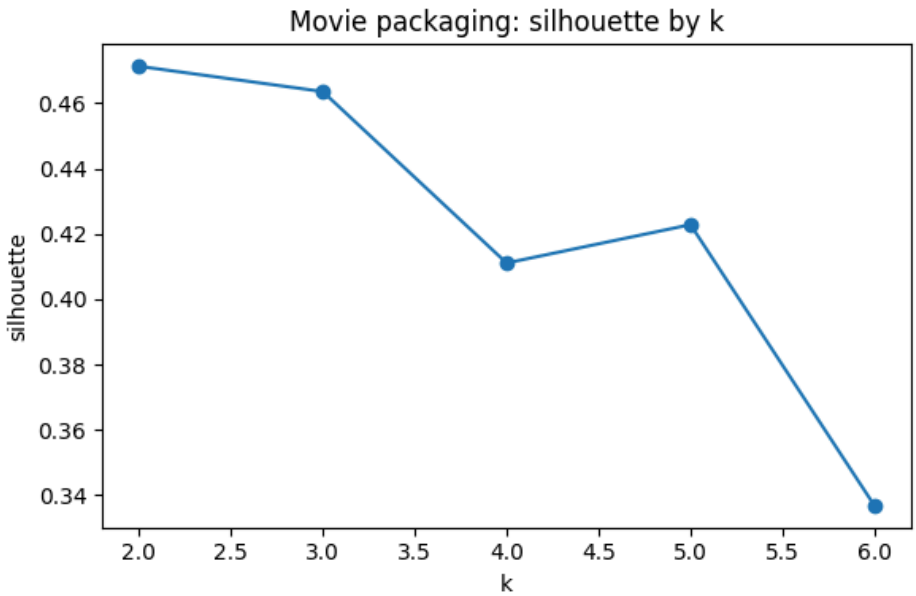


Figure 2. Spearman correlations between packaging measures and alternative star-proxy definitions remain small and near zero, indicating that the headline/inequality relationship is not

sensitive to reasonable proxy specification changes.

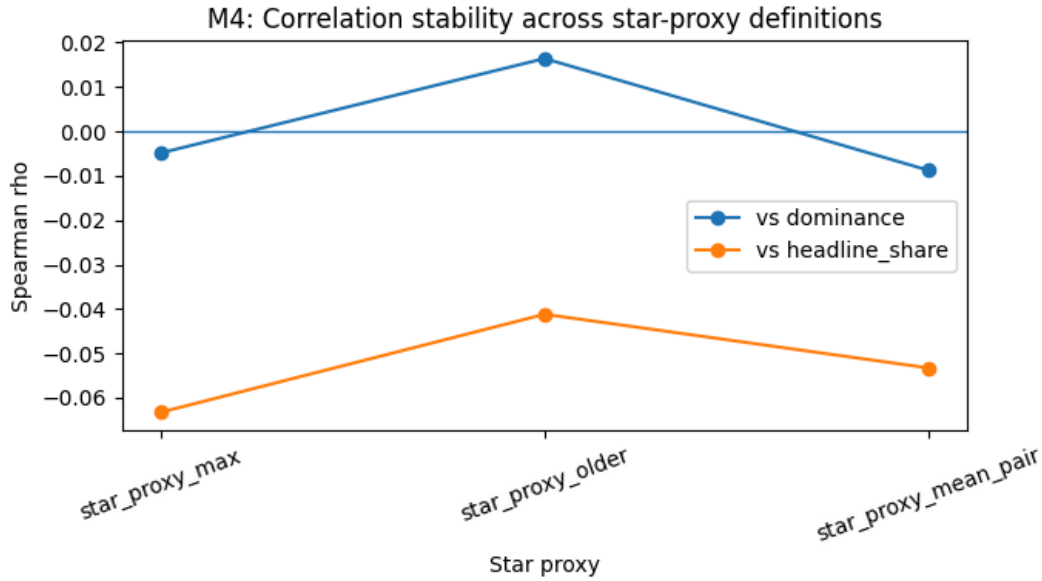


Table 1 – Early–late period splits (2005 cutoff) show small and statistically insignificant correlations across star-proxy definitions, indicating that the packaging–star relationship is temporally stable and not driven by a specific era.

Split year: 2005

period	count
early	113
late	111

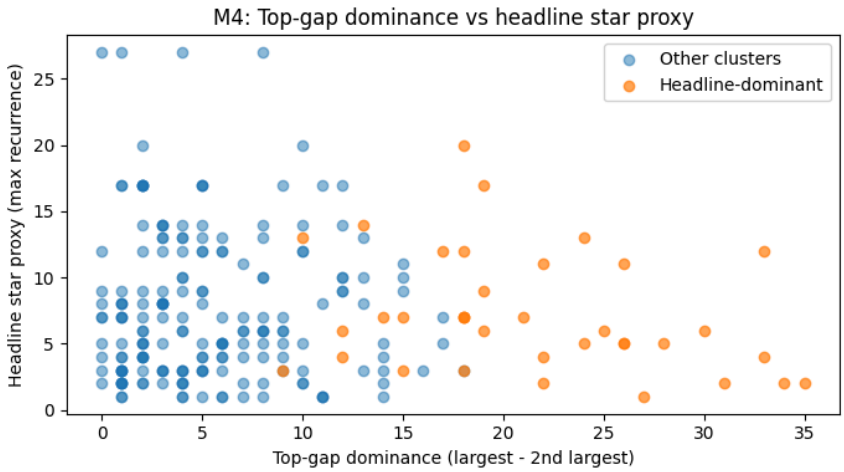
dtype: int64

	period	proxy	n_movies	rho_vs_dominance	p_vs_dominance	rho_vs_headline_share	p_vs_headline_share
0	early	star_proxy_max	113	-0.0794	0.4033	-0.1320	0.1634
1	early	star_proxy_older	113	-0.0933	0.3254	-0.1623	0.0859
2	early	star_proxy_mean_pair	113	-0.0824	0.3858	-0.1140	0.2293
3	late	star_proxy_max	111	0.0953	0.3200	0.0018	0.9850
4	late	star_proxy_older	111	0.1607	0.0919	0.1053	0.2714
5	late	star_proxy_mean_pair	111	0.0786	0.4125	-0.0161	0.8667

Table 2 – Bootstrap confidence intervals for headline vs. non-headline differences are centered at zero across star-proxy definitions, reinforcing the absence of a statistically meaningful packaging effect.

	proxy	median_diff_hd_minus_other	ci_low_95	ci_high_95
0	star_proxy_max	0.0	-2.00	2.0
1	star_proxy_older	0.0	-1.50	2.5
2	star_proxy_mean_pair	0.0	-1.25	1.5

Figure 4. The scatterplot shows no clear positive association between top-gap dominance and headline star-proxy concentration, indicating that films with larger internal age-gap inequality are not systematically driven by higher star recurrence.



Process Overview
A simple pipeline diagram/flow can show:

1. Raw dataset (age_gaps.csv)
2. Shared preprocessing (M3-compatible)
3. Rebuilda couple-level enriched table
4. Rebuild the multi-couple movie-level packaging table
5. Headline-couple extraction + star-proxy variants
6. Robustness tests:
    - proxy sensitivity
    - cluster comparison
    - temporal split
    - parity threshold sensitivity
    - bootstrap uncertainty
7. Output tables and plots for write-up

Use of Generative AI Tools
https://chatgpt.com/share/699d2b8c-b8f4-8010-98e2-8e1f88917390